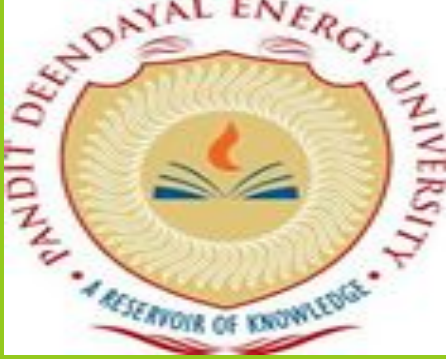




Brief Motivation

Banks and NBFCs outsource delinquent loan cases to collections agencies in monthly batches — often numbering in the thousands. Agencies must contact, follow up, and recover dues within a fixed cycle using a team of agents.

Without a centralized platform, this process is:

- Manual and unscalable across large case volumes
- Inconsistent in follow-up across calls, WhatsApp, and SMS
- Difficult to track in terms of agent productivity and recovery progress

A unified, automated collections platform directly improves contact rates, agent efficiency, and recovery outcomes within each monthly cycle.

Objective

- To develop a centralized case management system for agents to view and act on assigned collections cases
- To implement a bulk campaign module enabling structured outreach across thousands of cases per cycle
- To integrate AI-powered voice calling to maximize contact attempts beyond agent working hours
- To build multi-channel follow-up automation across WhatsApp and SMS
- To analyze customer repayment data to support case prioritization and load allocation

Methodology

Step 1: Domain & Codebase Understanding
Studied the collections domain — monthly case cycles, agent workflows, and recovery metrics. Explored the existing Spring Boot backend, React frontend (Umi + Ant Design Pro), Keycloak authentication, and Solr search infrastructure.

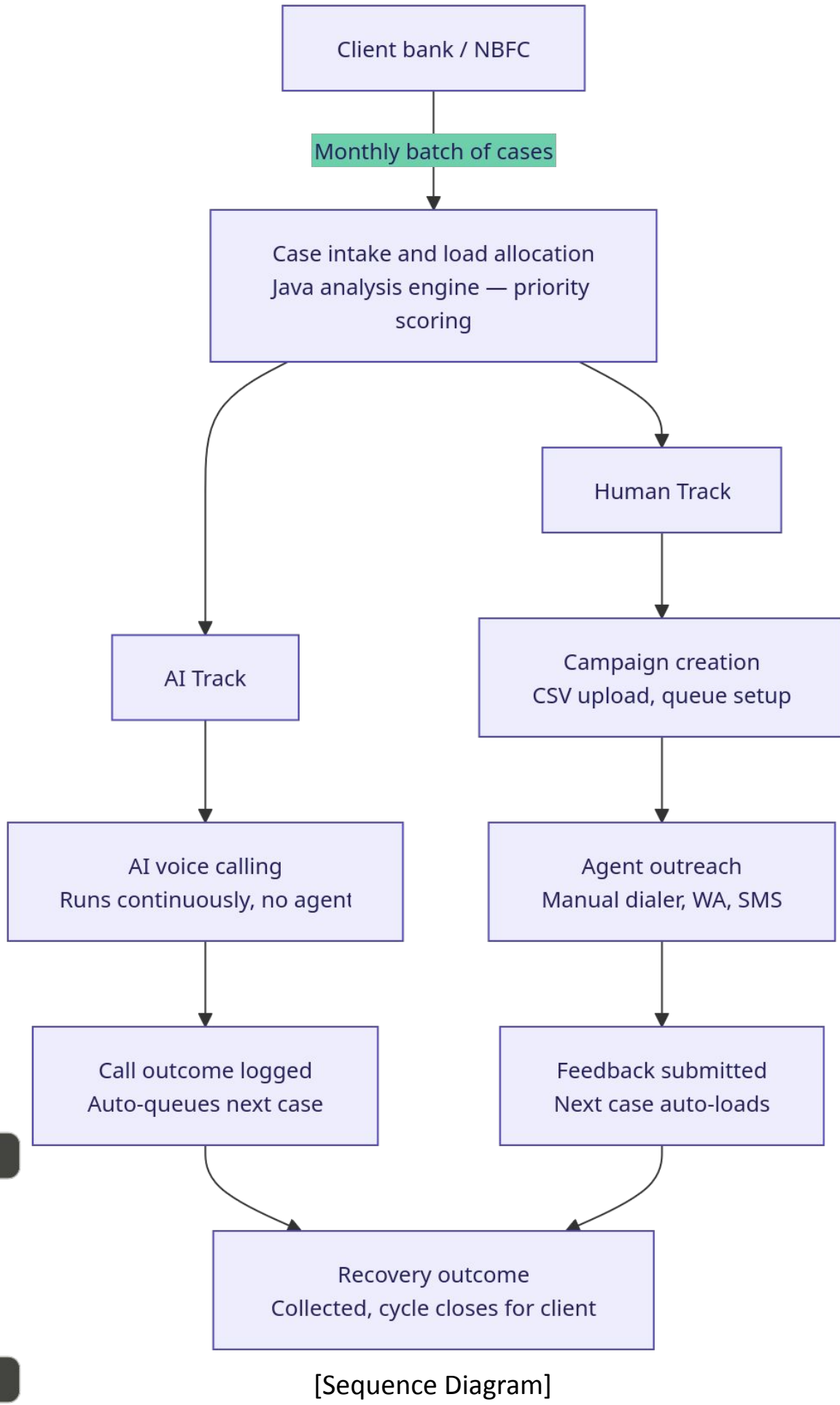
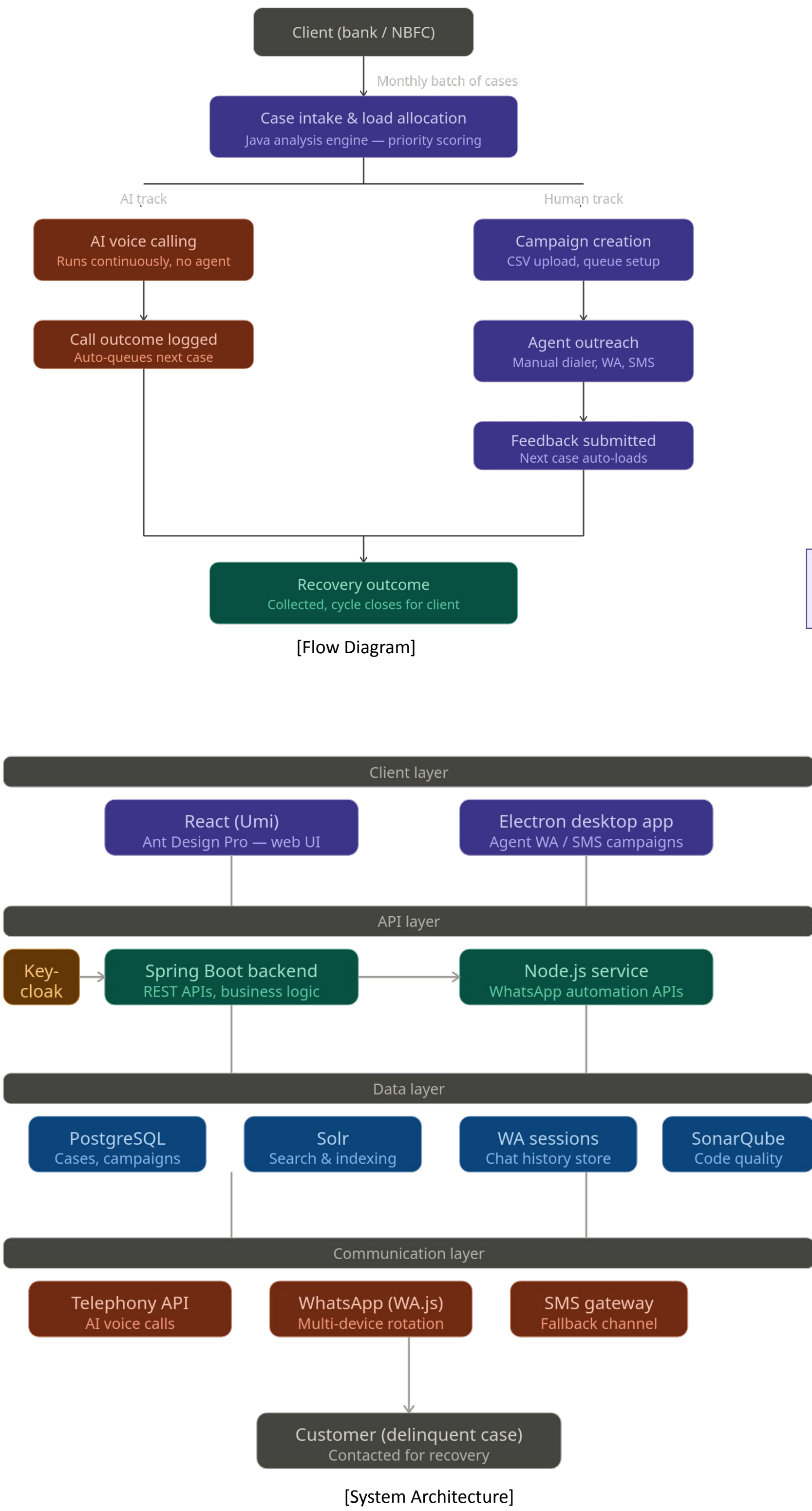
Step 2: Data Analysis & Case Prioritization
Analyzed 40,000+ customer records using Python (pandas). Examined overdue behavior, payment frequency, and recovery patterns. Generated summary reports and visualizations to support case load allocation.

Step 3: Core Platform Development
Built the case listing and case detail pages — integrated with paginated Spring Boot APIs. Displayed customer contact info, payment history, interaction notes, and status with real-time data fetching.

Step 4: Campaign Management System
Developed end-to-end campaign module — creation form, CSV upload with validation, backend parsing, preview step, and submission. Built campaign dashboard with real-time stats (calls completed, pending, success rate, amount collected).

Step 5: AI Voice Calling & Speed Dialer
Implemented speed dialer logic — on call end and feedback submission, system automatically loads and initiates the next case. Integrated with telephony backend for automated outbound calling.

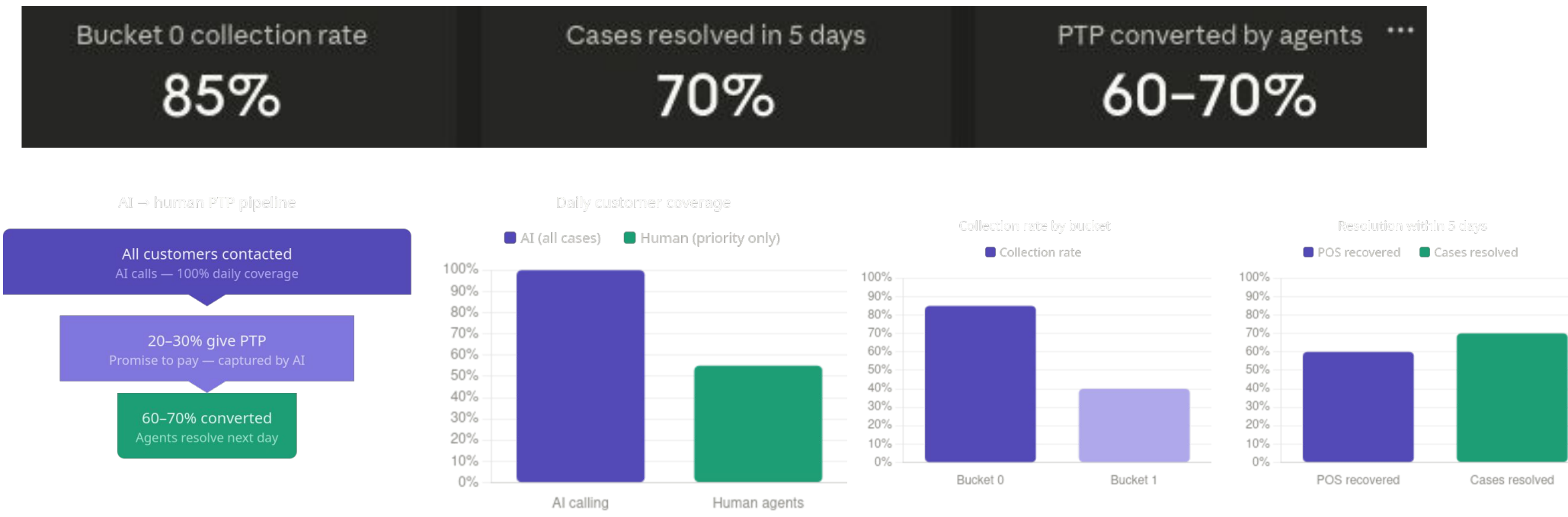
Step 6: Multi-Channel Outreach Automation
Built WhatsApp automation using Node.js — message templates, configurable delays, multi-device login, and automatic device rotation. Integrated SMS fallback on WhatsApp delivery failure. Packaged into an Electron desktop app for agent use.



Expected Results

- Increase overall contact rate by reaching customers beyond agent working hours through AI voice calling
- Enable structured bulk outreach via campaigns to reduce agent idle time and improve case throughput
- Provide a cost-effective alternative to scaling the agent workforce for high-volume collections
- Improve customer reachability through multi-channel outreach across voice, WhatsApp, and SMS
- Deliver real-time visibility into campaign progress, call outcomes, and recovery metrics through a unified dashboard
- Prioritise cases effectively using the Java-based analysis engine to maximise recovery within each monthly cycle

Outcomes



References

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